

Comparison of Untargeted Adversarial Attacks on CNN Classification Model

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Abstract: CNN (Convolutional Neural Networks) classification model [2] is an AI-driven model and is widely used in many streams such as autonomous systems, cybersecurity, and healthcare, due to its high accuracy in image classification. However, these models are vulnerable to adversarial attacks, where small perturbations in input data can cause the model to predict an incorrect class. There are many adversarial attacks available, such as targeted, untargeted, White-box attacks, and black-box attacks. Therefore, we aim to evaluate the comparison of untargeted attacks using FGSM (Fast Gradient Sign Method) and PGD (Projected Gradient Descent) [1] on the CNN Classification model by using the CIFAR-10 dataset. The key results show that the PGD attack is stronger; robustness depends on ϵ (epsilon) & training. FGSM is a time-efficient attack compared to PGD.

Keywords: CNN (Convolutional Neural Networks), Adversarial Attacks, FGSM (Fast Gradient Sign Method), and PGD (Projected Gradient Descent).

Introduction: Nowadays, Artificial Intelligence (AI) technologies are increasing rapidly in various fields. Particularly, CNN is vulnerable to adversarial attacks in high-stakes applications like healthcare, finance, and self-driving cars. These vulnerabilities have serious consequences. [4] To defend the ML systems, Adversarial attacks are performed.

An adversarial AI attack is a malicious technique that manipulates machine learning models by feeding them deceptive data to cause incorrect or unintended behavior. FGSM and PGD are the most common attacks in adversarial learning. Our goal is to compare these methods based on attack strength and stability on the CIFAR-10 dataset.

Background: Firstly, Convolutional Neural Networks – the backbone of Computer Vision research. CNN preserves the 2D nature of images and is capable of processing information. In this article, we are going to go through the CNN Classification tasks. Secondly, adversarial learning emerged from the early 2000s observation that machine learning models, particularly deep neural networks, were vulnerable to manipulate by small, carefully crafted input changes to make

model incorrect predictions. The first key paper on this topic appeared in 2004. The focus on finding vulnerabilities led to the development of various attacks, such as targeted attacks that force a specific incorrect output and untargeted attacks that do not worry about the output just make the model predict the incorrect classes. Our focus is to make a comparison of such untargeted methods, so we are using FGSM and PGD attacks methods. Which involves training the model on adversarial examples to improve robustness against them. Lastly, The FGSM and PGD both are the white-box attacks that means we know the architecture of the model and the input data and output. FGSM is a single step attack and was one of the first and most influential adversarial attacks, because it takes only one step to its less robust. PGD is a more powerful and refined extension of FGSM, its iteratively applies the FGSM step, but with the smaller step size, and projects the result back into a defined boundary to ensure the total perturbation remains small. By taking multiple steps, PGD explores the model's loss landscape more thoroughly, so it's powerful and effective.

Methodology: Mainly, these attacks algorithms are designed and operate in the white-box environment. The attacks are performed on simple CNN architecture for CIFAR-10 dataset, and the aim is to enforce the model to misclassify the adversarial image, regardless of the output. Although FGSM and PGD are routinely used to probe the model's robustness, it remains unclear when each attack should be preferred. This study systematically compares them on CIFAR-10 dataset to identify contexts where one attack provides stronger, more reliable evaluations than the other.

- 1. Dataset- CIFAR-10:** The CIFAR-10 dataset consists of 60000 colour images with 32x32x3 resolution size sorted into 10 classes (airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck), with 6000 images per class. The standard split is 50000 train and 10000 test images, class balanced. All the images are in the form of pixel and represent 32x32 pixel values for 3 different channels (RGB). For normalization, we use standard CIFAR-10 mean standard deviation.[\[3\]](#)
- 2. Model Architecture:** We use a compact CNN with two 3x3 convolutional layers 32 and 64 channels, and a common activation function Relu (Rectified Liner Unit). Each followed by 2x2 max-pooling, and two fully connected layers (4096 @ 512 @ 10). The network outputs class logits for the 10 CIFAR-10 categories.[\[2\]](#)

3. Training Setup: For training we use batch size of 64 train and 128 test. As a loss function we use cross-entropy and optimizer we use Adam with the learning rate of 1e-3. While for model performance we performed 20 training epochs cycles for preventing the overfitting.

4. Adversarial Methods (FGSM & PGD): We use normalization-aware stepping: inputs are standardized per channel using (m,s), while attack budget is ϵ and step sizes α are specified in pixel space and converted to normalized space by dividing by s per channel. FGSM applies one gradient-sign step within the ϵ -ball and the clip the result to the valid normalized range.

- $X_{adv} = \text{clip}_{[l,u]}(x + \epsilon \cdot \text{sign}(\nabla_x L))$ [1]

PGD perform iterative gradient-sign updates (10 steps) with step size $\alpha = 2/255$ in pixel unit converted to normalized, projecting back onto the ϵ -ball each iteration clipping to this formula.

- $[l,u]: x^{t+1} = \Pi_{B_\infty(x^0, \epsilon)}(x^t + \alpha \cdot \text{sign}(\nabla_x L))$ [1]

We condition on clean correctness (only samples correctly classified by the clean model are attacked) and, for each ϵ , report robust accuracy that means accuracy on adversarial examples, cross-entropy loss on adversarial logits, and attack success rate (ASR = 1- robust accuracy). We sweep $\epsilon \in \{0,2,4,8,12,16\} / 255$, with PGD using $\alpha = 2/255$ and 10 steps.

X – Clean normalized input image.

Y – Ground truth class label for x .

X_{adv} – Final adversarial example produce by attack.

X^0 – Starting point for PGD.

X^t – The adversarial iterate at step t of PGD; X^{t+1} is the next iterate.

L – Simplify the loss function Cross-entropy.

$\nabla_x L$ – Gradient of the loss w.r.t.

ϵ – Perturbation budget epsilon amount of noise.

α – Step size of PGD.

σ, μ – Per channel mean and standard deviation used for normalization.

$[l,u]$ – Valid range in normalized space corresponding to pixel range.

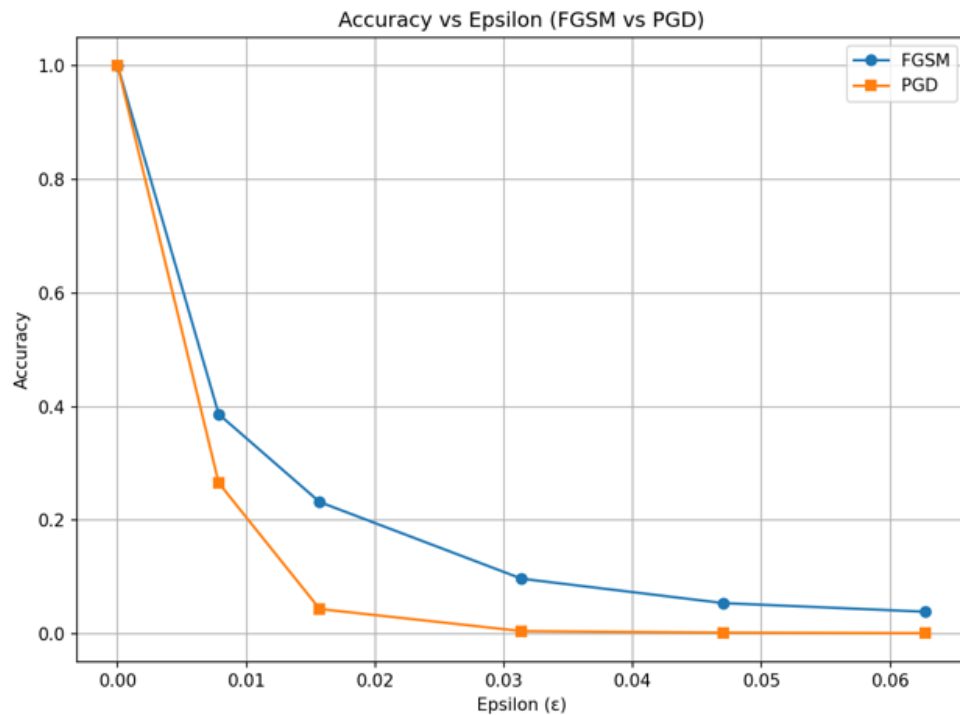
$B_\infty(x^0, \epsilon)$ – The ball of radius ϵ centered at x^0 .

$\Pi_{B_\infty(x^0, \epsilon)}$ – Projection operation that bring any point back to ball around the radius. [1]

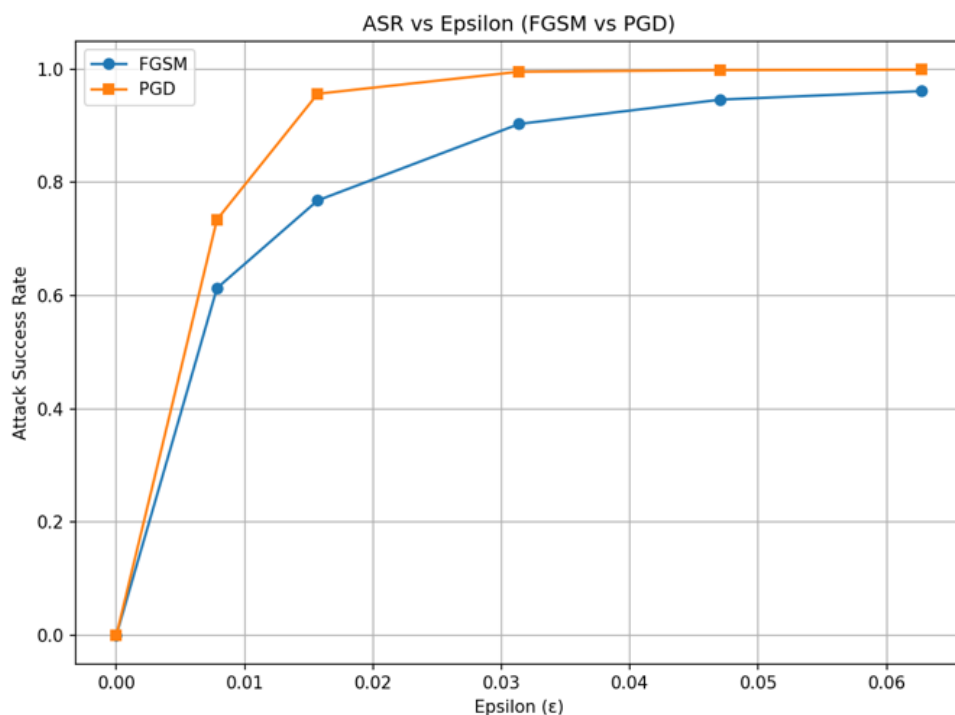
5. Results: Our primary finding shows that how epsilon effect on accuracy, loss and attack succession rate.

- **Accuracy vs Epsilon (FGSM & PGD):** Accuracy falls as epsilon grows, while PGD is consistently lower than FGSM at every epsilon. PGD accuracy is near zero, while FGSM still has

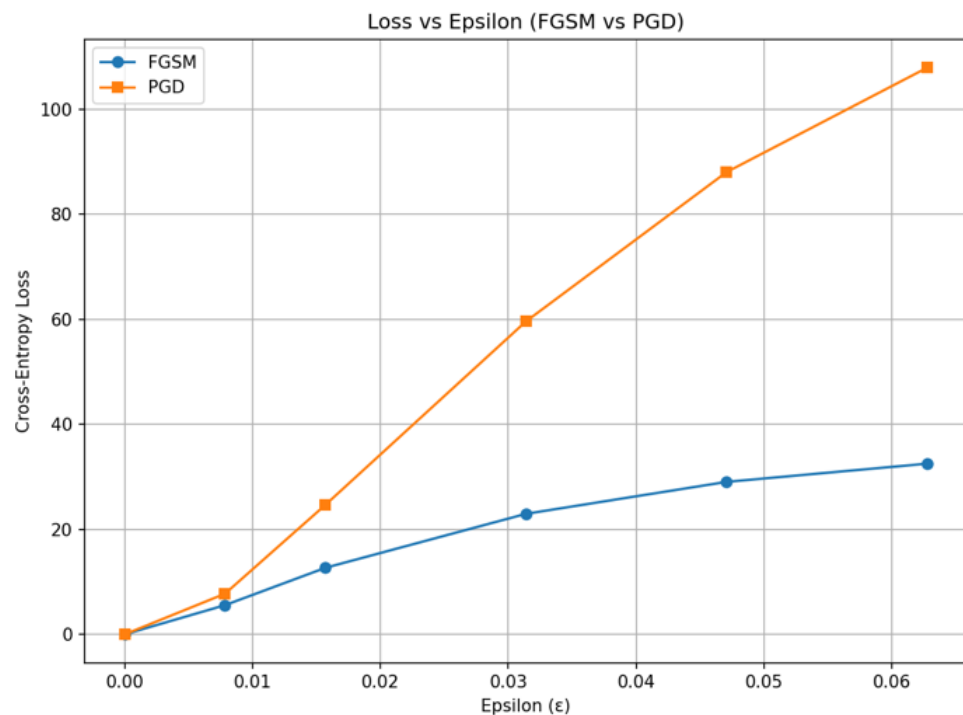
equivalent 0.23 so is low but not zero. whereas, By epsilon greater and equal to 0.03 PGD accuracy is equivalent to zero.



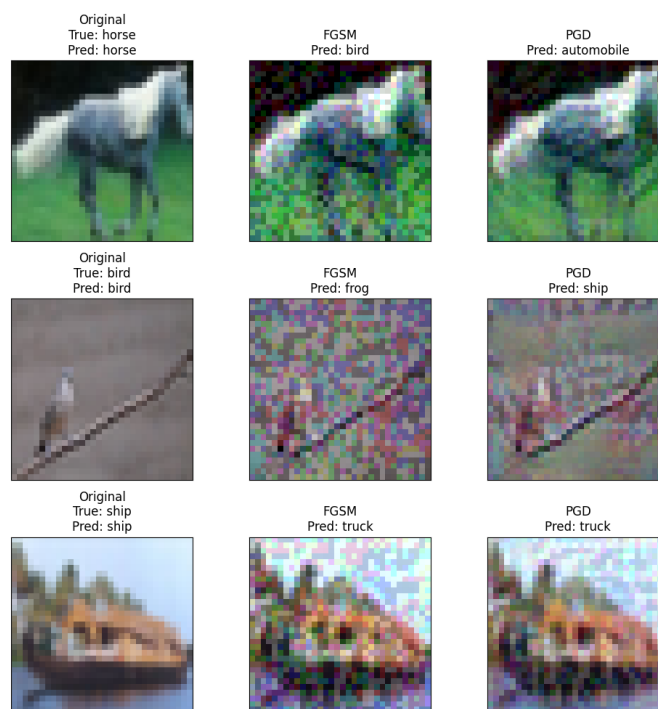
- **ASR vs Epsilon (FGSM vs PGD):** ASR rises with epsilon where PGD reaches equivalent to 1 much sooner at 0.03 epsilon and FGSM approaches approximately to 0.95 by epsilon of 0.06. So for achieving attack success rate FGSM required more epsilon



- Loss vs Epsilon (FGSM vs PGD):** Cross-entropy increases with the epsilon. PGD produced much higher losses than FGSM at matched epsilon. At the largest epsilon PGD is much greater than FGSM. So, it indicates stronger optimization pressure on the model.



- Sample Results of FGSM & PGD changes:**



6. Discussion: In practice, FGSM is a fast, one-step stress test that's sufficient for sanity checks, quick pipeline validation, coarse sensitivity screening, and low-compute demos especially at larger epsilon where even a single step often breaks a vanilla model. However, whenever conclusions matter evaluating a defense, reporting robustness numbers, training robust models, probing small budgets, or attempting targeted attacks multi-step PGD is necessary because its iterative optimization and projection more faithfully approximate the worst-case within the L_∞ ball and exposes gradient masking that FGSM can miss. A practical workflow is to start with normalization-aware FGSM to verify gradients and scaling, then escalate to PGD (e.g., 10–20 steps, $\alpha \approx 2/255$, random start) for results. If PGD's attack success rate markedly exceeds FGSM's at the same ϵ , treat FGSM as underestimating risk and rely on PGD. In short: use FGSM for quick checks; use PGD for credible robustness.

7. Conclusion: This study demonstrates that a standard CIFAR-10 CNN is highly sensitive to small, L_∞ -bounded perturbations and that attack strength scales sharply with epsilon. Practically, these findings imply that any deployment of vision models—even simple CNNs—requires defense-aware training and auditing. Robustness should be reported as a function of epsilon and attack configuration, not a single number. Looking forward, strengthening the baseline (e.g., ResNet/VGG), expanding to other norms (e.g., L_2), and incorporating adversarial training are natural next steps. Evaluating transferability, targeted variants, and data augmentation effects will further clarify robustness–accuracy trade-offs.

8. References:

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